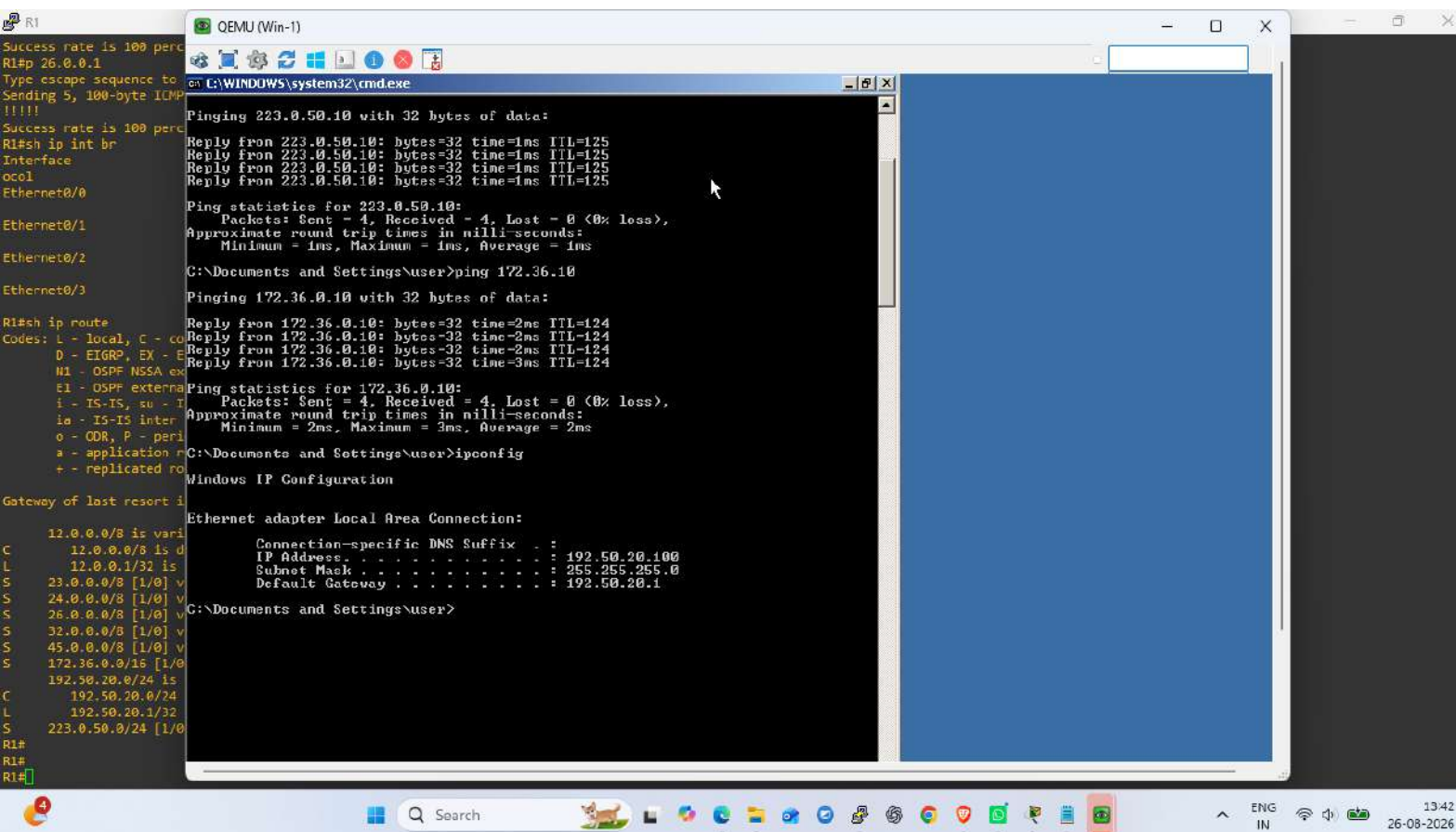
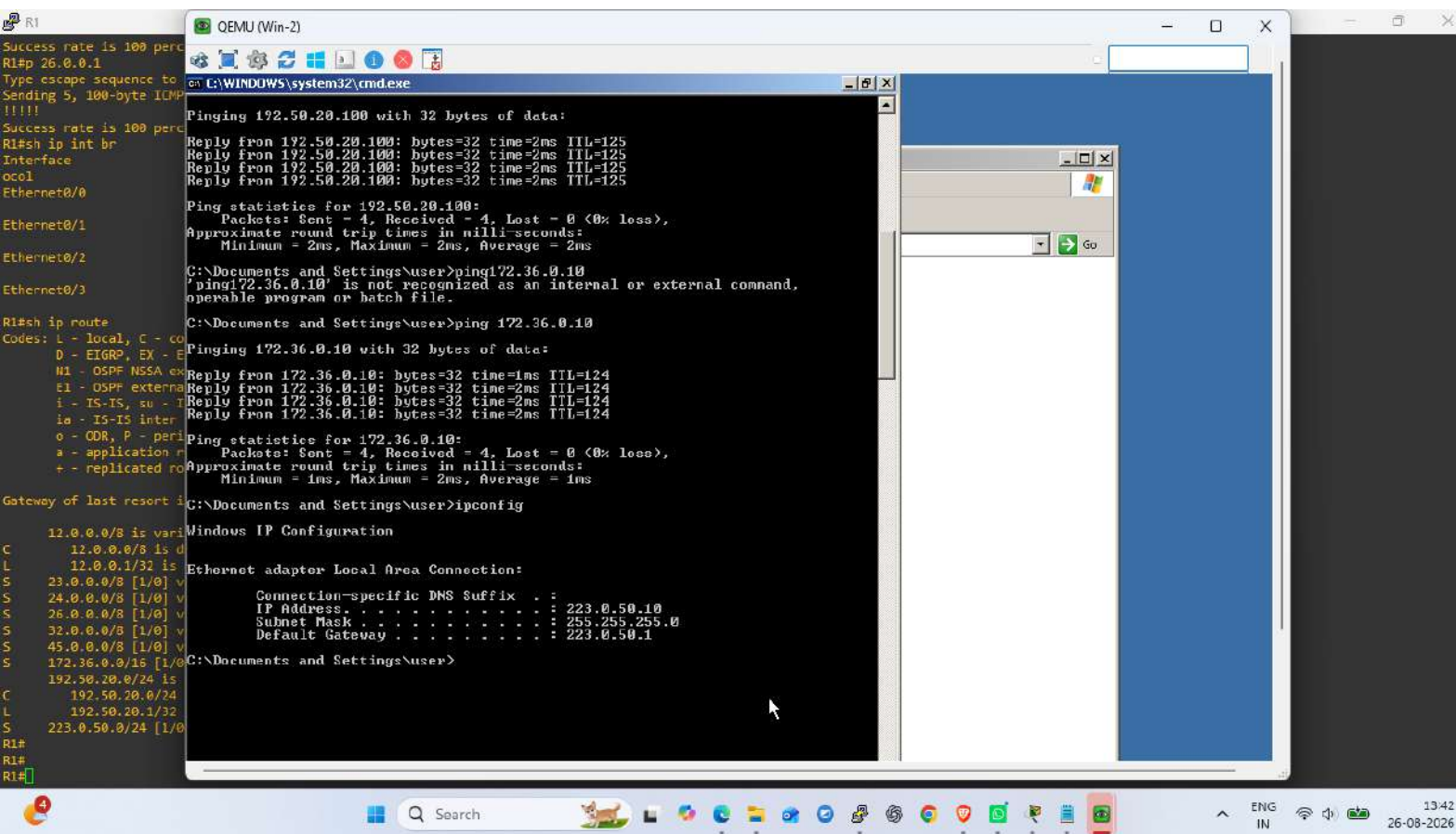
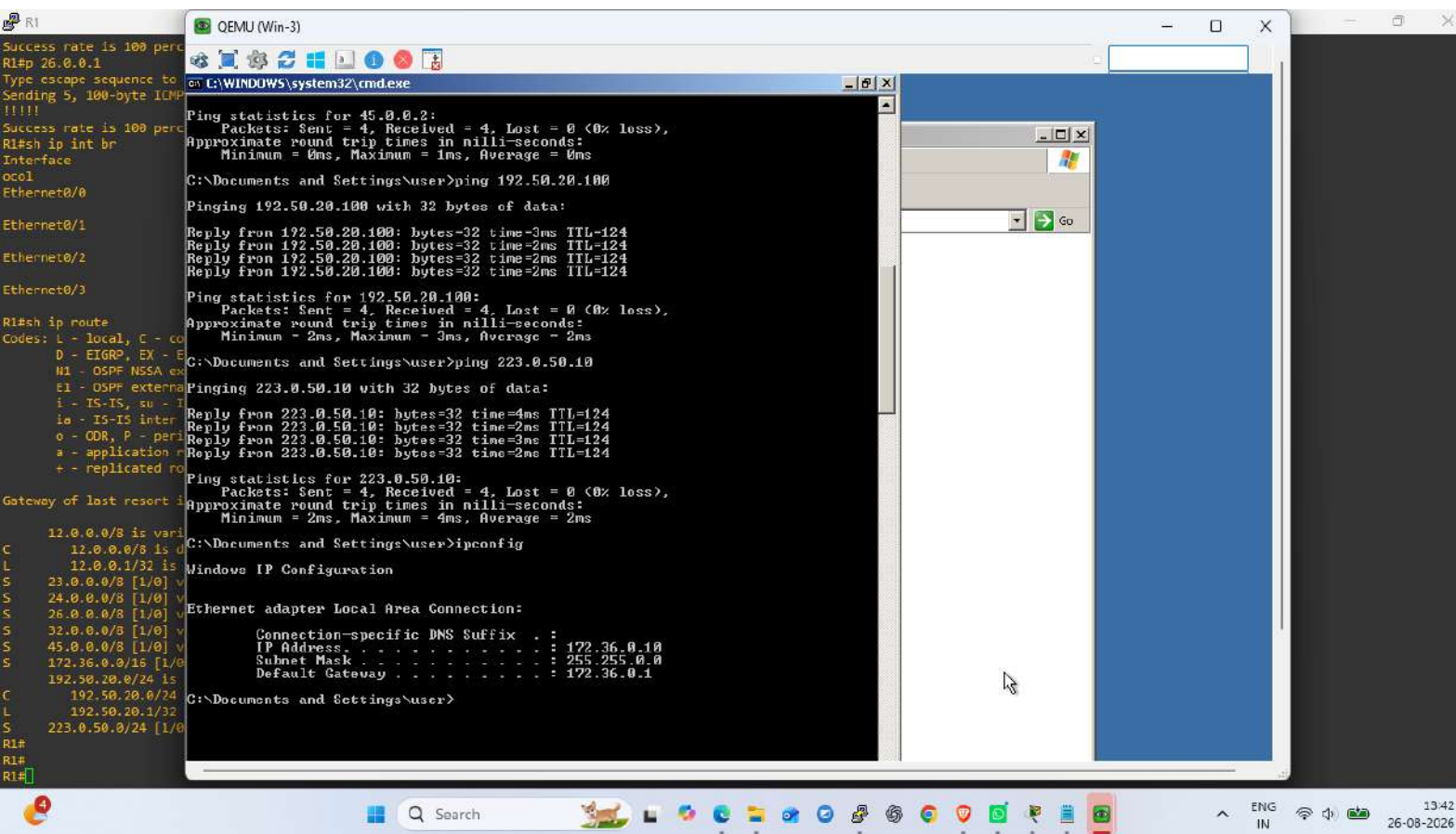








```
R1
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#p 26.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 26.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#sh ip int br
Interface IP-Address OK? Method Status Prot
o0/1
Ethernet0/0 192.50.20.1 YES manual up up
Ethernet0/1 12.0.0.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/1
L    12.0.0.1/32 is directly connected, Ethernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.50.20.0/24 is directly connected, Ethernet0/0
L    192.50.20.1/32 is directly connected, Ethernet0/0
S    223.0.50.0/24 [1/0] via 12.0.0.2

R1#
R1#
R1#
```

```
R2
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#p 12.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 12.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              12.0.0.2        YES manual up          up
Ethernet0/1              26.0.0.1        YES manual up          up
Ethernet0/2              23.0.0.1        YES manual up          up
Ethernet0/3              24.0.0.1        YES manual up          up
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Ethernet0/0
L    12.0.0.2/32 is directly connected, Ethernet0/0
23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/2
L    23.0.0.1/32 is directly connected, Ethernet0/2
24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    24.0.0.0/8 is directly connected, Ethernet0/3
L    24.0.0.1/32 is directly connected, Ethernet0/3
26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.1/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.30.0.0/16 [1/0] via 24.0.0.2
    [1/0] via 23.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2
R2#
R2#
```



```
R3
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#exit
R3#
*Aug 26 07:52:45.283: %SYS-5-CONFIG_I: Configured from console by console
R3#
R3#
R3#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              23.0.0.2        YES manual up          up
Ethernet0/1              32.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       +- replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
S    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 32.0.0.2
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1
R3#
R3#
R3#
R3#
R3#
```

```
R4
R4(config)#ip route 223.0.50.0 255.255.255.0 24.0.0.1
R4(config)#exit
R4#
*Aug 26 07:57:09.633: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int br
Interface                IP-Address      OK? Method Status  Protocol
Ethernet0/0              24.0.0.2        YES manual up      up
Ethernet0/1              45.0.0.1        YES manual up      up
Ethernet0/2              unassigned      YES TFTP  down    down
Ethernet0/3              unassigned      YES TFTP  down    down
R4#sh ip route
      ^
% Invalid input detected at '^' marker.

R4#
R4#sh ip rout
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       +- replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 45.0.0.2
    24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     24.0.0.0/8 is directly connected, Ethernet0/0
L     24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 45.0.0.2
    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     45.0.0.0/8 is directly connected, Ethernet0/1
L     45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1
R4#
R4#
R4#
R4#
```

```
R5
R5(config)#exit
R5#
*Aug 26 08:00:50.498: %SYS-5-CONFIG_I: Configured from console by console
R5#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              172.36.0.1      YES manual up          up
Ethernet0/1              32.0.0.2        YES manual up          up
Ethernet0/2              45.0.0.2        YES manual up          up
Ethernet0/3              unassigned      YES TFTP  down        down
R5#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              172.36.0.1      YES manual up          up
Ethernet0/1              32.0.0.2        YES manual up          up
Ethernet0/2              45.0.0.2        YES manual up          up
Ethernet0/3              unassigned      YES TFTP  down        down
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       +- replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
S    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     32.0.0.0/8 is directly connected, Ethernet0/1
L     32.0.0.2/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     45.0.0.0/8 is directly connected, Ethernet0/2
L     45.0.0.2/32 is directly connected, Ethernet0/2
S    172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     172.36.0.0/16 is directly connected, Ethernet0/0
L     172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
R5#
R5#
R5#
```



```
R6
R6(config)#
R6(config)#
R6(config)#exit
R6#
R6#
R6#
R6#
R6#
R6#
"Aug 26 08:04:23.795: %SYS-5-CONFIG_I: Configured from console by console
R6#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              223.0.50.1     YES manual up          up
Ethernet0/1              26.0.0.2       YES manual up          up
Ethernet0/2              unassigned     YES TFTP  down        down
Ethernet0/3              unassigned     YES TFTP  down        down
R6#
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        +- replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    223.0.50.0/24 is directly connected, Ethernet0/0
L    223.0.50.1/32 is directly connected, Ethernet0/0
R6#
R6#
R6#
R6#
```

